



RNCP35288 CDSD

Getaround Analysis

Deployment

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Contexte et enjeu



Réduire les frictions liées aux retards de restitution en définissant un délai minimum entre deux locations.



Optimiser la tarification des véhicules en recommandant un prix à partir des informations du véhicule.



Analyse des retards

57,53 %

De locations en retard

9 min

De retard médian

14,67 %

De prochaines locations impactés





Détermination du seuil optimal

Sans délai

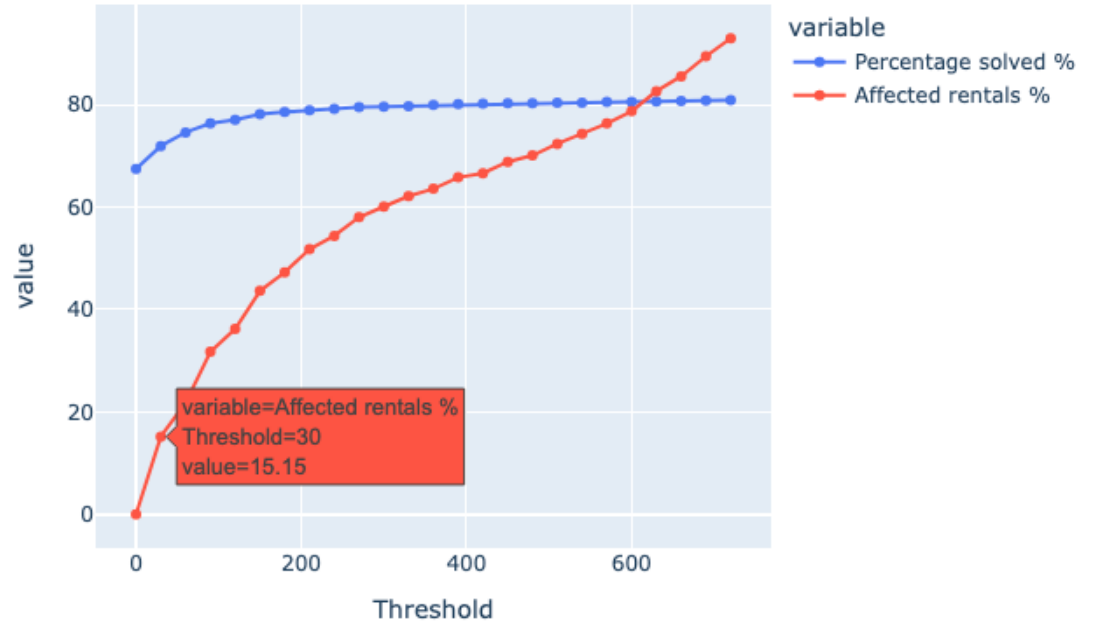
- 67 % problèmes évités

Avec un seuil de 30 min

- 72.42 % problèmes résolues (+5.4 points)
- 15.08 % locations consécutives impactées

Avec un seuil de 60 min

- +2,6 points de résolution
- 21.78 % locations consécutives impactées

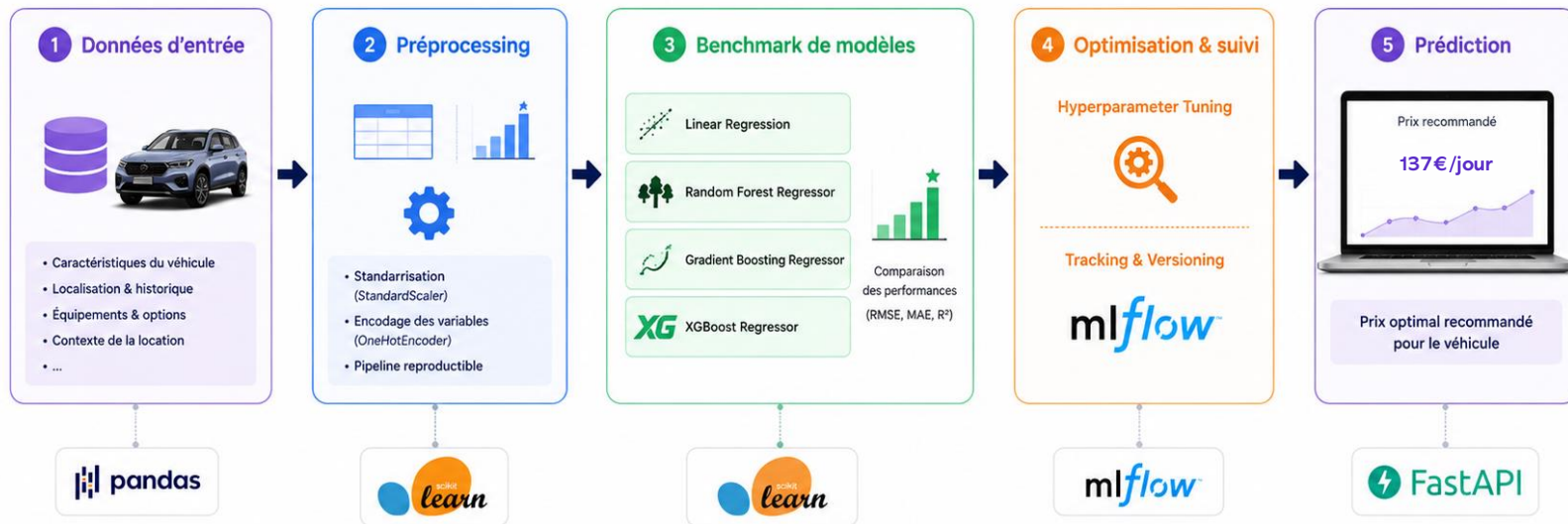




Machine Learning : prédiction du prix



[Lien Mlflow](#)



Objectif :

Recommander automatiquement le prix optimal d'une location à partir des caractéristiques du véhicule et du contexte.



Modèles comparés et optimisés



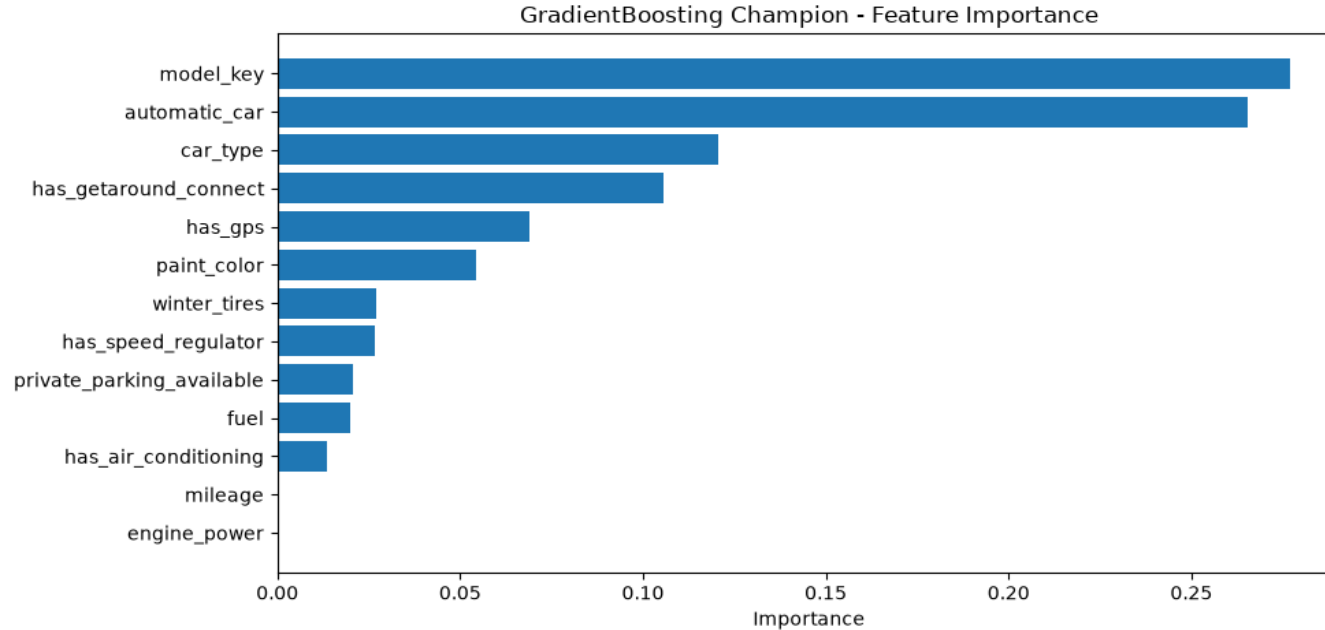
Suivi des modèles et reproductibilité



Déployable via API en production



Machine Learning : évaluation et interprétation



Metrics	Tune_GradientBoosting	Tune_XGBoost	XGBoost	GradientBoosting	RandomForest	LinearRegression
MAE	16.95	16.68	16.59	16.88	16.68	17.53
RMSE	23.91	23.92	24.13	24.15	24.51	24.94
R2	0.51	0.51	0.50	0.50	0.48	0.47



Mise en production



[Lien API](#)



[Lien Dashboard](#)



Getaround Rental Delay Analysis

Summary

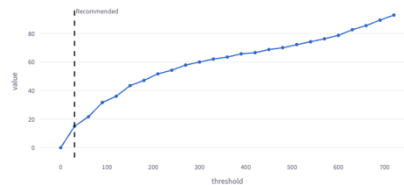
Recommended threshold: 30 minutes This threshold provides the best benefit-to-cost ratio. It solves approximately 72.42% of problematic situations while affecting only 15.08% of successive rentals.			
Threshold	Scope	Solved Cases	Affected Rentals
30 min	All Cars	72.42%	15.08%

1 How many rentals would be affected by the feature?

Affected Rentals
15.08%

A 30 minute buffer would affect 15.08% of successive rentals by making some booking combinations unavailable.

Affected Rentals vs Threshold



Getaround Pricing Simulator

Configure a vehicle and estimate its recommended daily rental price using the production Machine Learning model.

Vehicle Information

Vehicle Brand:

Fuel Type:

Paint Color:

Vehicle Type:

Mileage (km):

Engine Power (HP):

Vehicle Options

- ☒ Private Parking Available
- ☒ GPS
- ☒ Air Conditioning
- ☐ Automatic Transmission
- ☒ Getaround Connect
- ☒ Speed Regulator
- ☐ Winter Tires

Prediction completed successfully.

Recommended Daily Rental Price
137.49 €



Conclusion

Analyse métier

Seuil recommandé de **30 min** entre deux locations

Machine Learning

Recommandation de prix **en temps réel**

Outils métiers

Dashboard Analytics et simulateur de pricing

Industrialisation

MLflow • FastAPI • Docker • Hugging Face



Jedha

Merci pour votre attention
Des questions ?





API

Getaround Pricing API 1.0.0 OAS 3.1

openapi.json

Predict optimal rental prices for Getaround vehicles.

Champion model:
Tune_GradientBoosting

Registry:
getaround-pricing-model

Alias:
champion

default

GET / Root

GET /health Health

POST /predict Predict

Schemas

CarFeatures > Expand all object

HTTPValidationError > Expand all object

PredictionRequest > Expand all object

POST /predict Predict

Parameters

No parameters

Request body **required**

Example Value Schema

```
{
  "input": [
    {
      "model_key": "string",
      "mileage": 0,
      "engine_power": 0,
      "fuel": "string",
      "point_color": "string",
      "car_type": "string",
      "private_parking_available": true,
      "has_gps": true,
      "has_air_conditioning": true,
      "automatic_car": true,
      "has_getaround_connect": true,
      "has_speed_regulator": true,
      "winter_tires": true
    }
  ]
}
```

Responses

Code Description

200 Successful Response

Media type

application/json

Controls Accept header:

Example Value Schema

Curl

```
curl -X 'POST' \
  'https://stoneray-fastapi-getaround.hf.space/predict' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "input": [
      {
        "model_key": "BMW",
        "mileage": 10000,
        "engine_power": 200,
        "fuel": "petrol",
        "point_color": "white",
        "car_type": "SUV",
        "private_parking_available": true,
        "has_gps": true,
        "has_air_conditioning": true,
        "automatic_car": true,
        "has_getaround_connect": true,
        "has_speed_regulator": true,
        "winter_tires": true
      }
    ]
  }'
```

Request URL

https://stoneray-fastapi-getaround.hf.space/predict

Server response

Code Details

200

Response body

```
{
  "prediction": [
    131.4215113091957
  ]
}
```

Response headers

```
access-control-allow-origin: https://stoneray-fastapi-getaround.hf.space
access-control-expose-headers: *
content-length: 35
content-type: application/json
date: Thu, 25 Jul 2024 13:43:49 GMT
link: <https://huggingface.co/spaces/stoneray/FASTAPI_getaround>;rel="canonical"
server: uvicorn
vary: origin,access-control-request-method,access-control-request-headers
x-praised-host: http://10.111.171.230
x-praised-path: /predict
x-praised-replica: vk7qhr6-cmmt
x-request-id: xzsQux
```



Download

Call

No links



Ressources projet

Github repository : https://github.com/Manjakaranja/Getaround_project_2026

Dashboard : https://huggingface.co/spaces/stoneray/STREAMLIT_Getaround

API : <https://stoneray-fastapi-getaround.hf.space/docs>

Mlflow : <https://stoneray-mlflow-getaround.hf.space/#/>